

$\mathcal{K}_{0^+}^{\#1}$		$\mathcal{K}_{0^+}^{\#1\alpha\beta\chi}$					
$\mathcal{K}_{0^+}^{\#1\dagger}$	$-\frac{3\gamma_1 k^2}{2}$	0					
$\mathcal{K}_{0^+}^{\#1\dagger\alpha\beta\chi}$	0	$3k^2\overset{(4)}{\mathcal{K}}_{15}+3\overset{(2)}{\mathcal{K}}_1$	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1\alpha}$	$\mathcal{K}_{1^+}^{\#2\alpha}$	
$\mathcal{K}_{1^+}^{\#1\dagger\alpha\beta}$	$\gamma_2 k^2+2\overset{(2)}{\mathcal{K}}_1$	$\frac{\gamma_2 k^2+2\overset{(2)}{\mathcal{K}}_1}{\sqrt{2}}$	0	0			
$\mathcal{K}_{1^+}^{\#2\dagger\alpha\beta}$	$\frac{\gamma_2 k^2+2\overset{(2)}{\mathcal{K}}_1}{\sqrt{2}}$	$\frac{1}{2}(\gamma_2 k^2+2\overset{(2)}{\mathcal{K}}_1)$	0	0			
$\mathcal{K}_{1^+}^{\#1\dagger\alpha}$	0	0	$-k^2\overset{(4)}{\mathcal{K}}_2$	$-\sqrt{2}k^2\overset{(4)}{\mathcal{K}}_2$			
$\mathcal{K}_{1^+}^{\#2\dagger\alpha}$	0	0	$-\sqrt{2}k^2\overset{(4)}{\mathcal{K}}_2$	$-2k^2\overset{(4)}{\mathcal{K}}_2$	$\mathcal{K}_{2^+}^{\#1\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1\alpha\beta\chi}$	
					$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta}$	$-9k^2\overset{(4)}{\mathcal{K}}_2$	0
					$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$					
$\mathcal{J}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0					
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{3(k^2 \overset{(4)}{\kappa}_{15} + \overset{(2)}{\kappa}_1)}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	
	$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{2}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0		
	$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{1}{3 \text{Det}(1^+)}$	0	0		
	$\mathcal{J}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-\frac{1}{9 k^2 \overset{(4)}{\kappa}_2}$	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_2}$		
	$\mathcal{J}_{1^+}^{\#2} \dagger^\alpha$	0	0	$-\frac{\sqrt{2}}{9 k^2 \overset{(4)}{\kappa}_2}$	$-\frac{2}{9 k^2 \overset{(4)}{\kappa}_2}$	$\mathcal{J}_{2^+}^{\#1}$	$\mathcal{J}_{2^+}^{\#1}$
				$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{1}{9 k^2 \overset{(4)}{\kappa}_2}$	0	
				$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

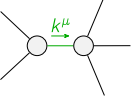
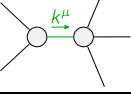
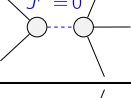
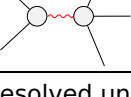
Abbreviations used in matrices

$$Y_1 == \overset{(4)}{\mathcal{K}}_1 + 7 \overset{(4)}{\mathcal{K}}_2 \ \&\& \ Y_2 == 2 \overset{(4)}{\mathcal{K}}_{15} + 9 \overset{(4)}{\mathcal{K}}_2 - \overset{(4)}{\mathcal{K}}_3 \ \&\&$$

$$\text{Det}(0^+) == -\frac{3}{2} k^2 (\overset{(4)}{\mathcal{K}}_1 + 7 \overset{(4)}{\mathcal{K}}_2) \ \&\& \ \text{Det}(1^+) == \frac{3}{2} k^2 (2 \overset{(4)}{\mathcal{K}}_{15} + 9 \overset{(4)}{\mathcal{K}}_2 - \overset{(4)}{\mathcal{K}}_3) + 3 \overset{(2)}{\mathcal{K}}_1$$

Lagrangian

$$\begin{aligned} &\overset{(2)}{\mathcal{K}}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\mathcal{K}}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\mathcal{K}}_1 \partial_\beta \mathcal{K}^\delta_{\ \chi\delta} \partial^{\chi} \mathcal{K}^{\alpha\ \beta}_{\ \alpha} + \\ &\overset{(4)}{\mathcal{K}}_2 \partial_\chi \mathcal{K}^\delta_{\ \beta\delta} \partial^{\chi} \mathcal{K}^{\alpha\ \beta}_{\ \alpha} + \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \alpha\chi} + 18 \overset{(4)}{\mathcal{K}}_2 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \beta\chi} - 2 \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \beta\chi} + \\ &18 \overset{(4)}{\mathcal{K}}_2 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \chi\alpha} - \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \chi\alpha} - 6 \overset{(4)}{\mathcal{K}}_2 \partial^{\chi} \mathcal{K}^{\alpha\ \beta}_{\ \alpha} \partial_\delta \mathcal{K}^{\delta}_{\ \chi\beta} + \\ &\frac{9}{2} \overset{(4)}{\mathcal{K}}_2 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \beta\chi} - \frac{1}{2} \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\ \beta\chi} + \overset{(4)}{\mathcal{K}}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\mathcal{K}}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\ \chi}_{\ \beta} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\ \beta} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^{\chi} \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^{\chi} \partial^\alpha \mathcal{J}^{\delta\beta}_{\ \delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^{\chi} \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^{\chi} \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_{\ \delta}{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha}_{\ \delta} ==$ $3 \partial_\epsilon \partial_\delta \partial^{\chi} \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^{\chi} \partial^\beta \mathcal{J}^{\delta\alpha}_{\ \delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^{\chi} \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_{\ \delta}{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta}_{\ \delta}$	
Total # constraint(s):	11		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{3+2 \sqrt{17}}{\overset{(4)}{\mathcal{K}}_2}$
	1	0	$\frac{3-2 \sqrt{17}}{\overset{(4)}{\mathcal{K}}_2}$
	1	$-\frac{\overset{(2)}{\mathcal{K}}_1}{\overset{(4)}{\mathcal{K}}_{15}}$	$-\frac{1}{3 \overset{(4)}{\mathcal{K}}_{15}}$
	3	$\frac{2 \overset{(2)}{\mathcal{K}}_1}{-2 \overset{(4)}{\mathcal{K}}_{15} - 9 \overset{(4)}{\mathcal{K}}_2 + \overset{(4)}{\mathcal{K}}_3}$	$\frac{2}{6 \overset{(4)}{\mathcal{K}}_{15} + 27 \overset{(4)}{\mathcal{K}}_2 - 3 \overset{(4)}{\mathcal{K}}_3}$
Resolved unitarity condition(s):	(Demonstrably impossible)		